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Advances and Trends in Artificial Intelligence

Theory and Applications

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Preface

In the last few years, significant progress has been made in computing infrastructures, deep learning, and big foundation models, catalyzing the rapid advancement of artificial intelligence. Computing infrastructures have witnessed remarkable improvements, with the popularity of cloud computing and graphical processing units providing researchers and practitioners with immense computational resources to train and deploy complex AI models. Deep learning enables machines to automatically learn intricate patterns and representations from vast amounts of data. Additionally, big foundation models, such as transformer architectures, have emerged as building blocks for natural language processing and computer vision, empowering AI systems with enhanced understanding and decision-making capabilities. These developments have accelerated the development of more effective intelligent systems to solve real-world complex problems. Moreover, innovative applications of artificial intelligence are continually emerging.

This volume contains the proceedings of the 36th edition of the International Conference on Industrial, Engineering, and other Applications of Applied Intelligent Systems (IEA/AIE 2023), which was held on July 19–22, 2023, in Shanghai, China. IEA/AIE is a yearly conference that focuses on applications of applied intelligent systems to solve real-life problems in all areas including business and finance, science, engineering, industry, cyberspace, bioinformatics, automation, robotics, medicine and biomedicine, and human-machine interactions. IEA/AIE 2023 was organized in cooperation with the ACM Special Interest Group on Artificial Intelligence (SIGAI). This year, 129 submissions were received. Each paper was evaluated using double- blind peer review by at least three reviewers from an international Program Committee consisting of 174 members from 25 countries. Based on the evaluation, a total of 50 papers were selected as full papers and 20 as short papers, which are presented in the two volumes of this book. The acceptance rate was 54%. We would like to thank all the reviewers for the time spent on writing detailed and constructive comments for the authors, and to the latter for the proposal of many high-quality papers.

In the program of IEA/AIE 2023, four special sessions were organized: Collective Intelligence in Social Media (CISM 2023), Intelligent Knowledge Engineering in Decision Making Systems (IKEDMS 2023), Intelligent Systems and e-Applications (iSeA 2023) and Causal Inference and Machine Learning (CIML 2023). In addition, two keynote talks were given by two distinguished researchers, one by Michael Sheng from Macquarie University (Australia) and the other by Cewu Lu from Shanghai Jiaotong University (China). We would like to thank everyone who has contributed to the success of this year's edition of IEA/AIE, that is the authors, reviewers, keynote speakers, Program Committee members and organizers. We would like thank all the sponsors and institutions that have provided strong support for the conference, including Association for the Advancement of Artificial Intelligence (AAAI), Association for Computing Machinery (ACM), Shanghai Computer Society, i-SOMET Incorporated Association, Springer and the Conference Management Toolkit. We are particularly grateful to the organizers of

this year's event, including Shanghai University of Finance and Economics, Fudan University and Shanghai Business School. Their support was crucial for the success of this year's conference.

May 2023

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Keynote Speech

Smart IoT Sensing for Aging Well: Research Activities and Future Directions

Michael Sheng

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Abstract. Worldwide, the population is aging due to increasing life expectancy and decreasing fertility. The significant growth in older population presents many challenges to health and aged care services. Over the past two decades, the Internet of Things (IoT) has gained significant momentum and is widely regarded as an important technology to change the world in the coming decade. Indeed, IoT will play a critical role to improve productivity, operational effectiveness, decision making, and to identify new business service models for social and economic opportunities. Indeed, with the development of low-cost, unobtrusive IoT sensors, along with data analytics and artificial intelligence (AI) technologies, there is now a significant opportunity to improve the wellbeing and quality of life particularly of our older population. In this talk, we will overview some related research projects and also discuss several research directions.

Biography: Michael Sheng is a full Professor and Head of School of Computing at Macquarie University, Sydney, Australia. Before moving to Macquarie University, he spent 10 years at School of Computer Science, the University of Adelaide. Michael Sheng's research interests include the Internet of Things (IoT), service computing, big data analytics, machine learning, and Web technologies. He is ranked by Microsoft Academic as one of the Most Impactful Authors in Services Computing (ranked Top 5 All Time) and in Web of Things (ranked Top 20 All Time). Michael Sheng is the recipient of AMiner Most Influential Scholar in IoT (2018), ARC (Australian Research Council) Future Fellowship (2014), Chris Wallace Award for Outstanding Research Contribution (2012), and Microsoft Research Fellowship (2003). He is the Vice Chair of the Executive Committee of the IEEE Technical Community on Services Computing

(IEEE TCSVC), the Associate Director of Macquarie University Smart Green Cities Research Center, and a member of the ACS (Australian Computer Society) Technical Advisory Board on IoT.

Behavior Understanding and Embodied Intelligence

Cewu Lu

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Abstract. This talk discusses the problem of behavior understanding of intelligent agents. From the perspective of machine cognition, how to make the machine understand the behavior? We introduce the work of human behavior knowledge engine and behavior semantic unification under Poincaré space. From the perspective of neurocognition, what is the inner relationship between machine semantic understanding and brain neurocognition? We introduce how to explain the intrinsic relationship between visual behavior understanding and brain nerves, and establish a stable mapping model. From the perspective of embodied cognition, how to make the robot have the first-person behavior ability? We introduce the proposed PIE (perception-imagination-execution) scheme, in which the representative work grassNet reaches the human level for the first time in grasping unknown objects.

Biography: Cewu Lu is a professor of Shanghai Jiao Tong University. In 2016, he was selected under the National "1000 Youth Talents Plan". In 2018, he was selected as one of 35 Innovators Under 35 (MIT TR35) by MIT Technology Review. In 2019, he was awarded Qiu Shi Outstanding Young Scholar. In 2020, he was awarded the Special Prize of Shanghai Science and Technology Progress Award (ranked third). In 2021, he won the title of Highly Cited Scholar in China. In 2022, he was awarded one of the best papers in IROS (6/3579). he, as the corresponding author or the first author, has published 100 papers in high-level journals and conferences. He has served as reviewer for Science main issue, Nature sub-journal, Cell sub-journal and other journals, area chair of NeurIPS, CVPR, ICCV, ECCV, IROS, ICRA. His research interests fall mainly in Computer Vision and Robot Learning.

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